STATEMENT OF INTENT

"Where do we stand amidst the well-defined laws and precise mathematical order that governs our Universe"; a question that has often intrigued me. Since 2012, when the Higgs Boson was discovered at the Large Hadron Collider (LHC), I have found myself immersed in reading about the Standard Model (SM). My self-study efforts, albeit at a fundamental level, opened the gate of a new world- the one of Particle Physics. In order to delve into the mysterious and immensely fascinating world of particles, I am applying to the PhD program at the University of Maryland.

Driven by the urge to understand more about the laws of Physics that govern our everyday lives as much as they govern the millions of stars that lie billions of light years away from us, in July 2015, I signed up for B. Tech. in Engineering Physics at *Delhi Technological University* (DTU). This research-oriented course at DTU was the perfect opportunity to explore different branches of Physics, such as Atomic and Molecular Physics, Optics, and simultaneously, receive hands-on exposure that has taken me beyond confines of our prescribed syllabus. In August 2017, as a prerequisite for my research project, I began studying the application of Quantum Mechanics in Condensed Matter Physics. This involved understanding how Density Functional Theory (DFT) is employed to solve the quantum-many body problem, along with approximation techniques proposed before DFT. I spent the next few months going through literature and building a basic understanding of the subject. However, balancing this additional work while simultaneously trying to meet rigors of my university studies, was immensely challenging. It was only through patience and perseverance that I was able to acquire the required knowledge to begin my research project. In December 2017, I began my research work in Heusler alloys wherein we use the full-potential DFT based calculations, as implemented by the WIEN2k Package.

I have authored two research papers in this domain, which I recently presented at the *Recent Advances in Condensed Matter Physics* conference (Kurukshetra University, India). I am the first author of the paper, "Effect of disorders on Half-Metallic Ferromagnetism in Cr_2CoAl inverse Heusler alloy" in which we have simulated five types of structural disorders on the alloy and found its spin polarization to check its viability for use in Spintronics. I also co-authored the paper, "Study of $FeCrSn_{1-x}Ga_x$ Heusler Alloys: Tuning Fermi Level to attain Half-Metallic Ferromagnetism". We doped the compound $FeCrSn$ with different concentrations of Ga and found the resulting spin polarization of the compound. Our current work involves reducing the disorder concentration and employing the modified Becke-Johnson exchange potential with the local density approximation for the calculations of structural disorders in Cr_2CoAl. I have thoroughly enjoyed my work in the field of computational condensed matter physics.

However, my long-standing interest in Particle Physics paved the way for my recently concluded internship under the 'Summer Research Fellowship Programme' at the Indian Institute of Technology Delhi wherein I studied the basic concepts of the Standard Model of Particle Physics. Even though this internship involved the introduction of concepts that one usually encounters during graduate courses, I wanted to explore more about this subject, including any experiential learning opportunities I could gain before embarking on the doctorate program. It is due to my mentor, Dr. Pradipta Ghosh, who constantly focused on strengthening my basic concepts and making me understand the topics in depth, that I was able to appreciate the aesthetics of the SM. It was during the numerous discussion sessions with my mentor in which he elaborated upon the significance of beyond the Standard Model (BSM) physics, such as Supersymmetry (SUSY). The ongoing searches for BSM at the LHC as well as other experiments, e.g. Dark matter searches, are continuously looking for signs of superpartners and hence the BSM Physics. This thrilling ambience of High Energy Physics, that is the unavoidable necessity of BSM Physics to overpower the shortcomings of the SM and simultaneously the hitherto unseen experimental signals of BSM Physics, motivates me to contribute meaningfully in this subject area.

In order to explore other exciting branches of Physics, I volunteered to carry out remote computing work for the ALTAIR project under Dr. Justin Albert (University of Victoria). Here, I was required to test the high-altitude balloon flight path prediction software. This experience helped me understand the importance of strengthening my programming skills, and I began coding in Python so that I could have a strong hold over the language before starting the doctorate program. This project along with my other academic experiences have prepared me for a future in the tough regime of Particle Physics by honing key skills like critical thinking and problem solving and instilling in me qualities like persistence and not giving up in the face of challenges. With this attitude, I was able to secure the first position in the last three semesters in my department, and I currently hold the third overall rank in the ongoing bachelor's program.

A major factor behind applying to the Department of Physics at UMD is the wide variety of courses offered, that would not only help me strengthen my concepts in theoretical particle physics but would also introduce me to the experimental aspect of it. I am particularly interested in taking the courses 'Elementary Particle Physics', 'Advanced Quantum Field Theory' and 'Beyond the Standard Model.' I also look forward to being part of the *Elementary Particle Physics- Theory* group at the *Maryland Center for Fundamental Physics* and I would be honored to work under the mentorship of Prof. Raman Sundrum, whose previous work has provided a possible solution to the Hierarchy problem. As his research interests lie in a variety of fields in the BSM physics, I believe that with his guidance I will be able to contribute constructively to this field. I am also interested in working under the supervision of Prof. Rabindra Mohapatra and Prof. Zackaria Chacko, as their research fields align with my interests.

Pursuit of research, and a subsequent career in academia, is how I envision myself after this PhD. I believe that the immense opportunities at UMD will help me understand the subject in its entirety, and where my academic vision will come to fruition. A desire to understand not just the laws of Physics, but also being able to answer the bigger questions about the genesis of this universe captures my interest.